

Question	Marking Guidance	Mark	Comments
04.1	(Locus) 1. (Fixed) position of a gene on DNA (molecule)/chromosome; (Exon) 2. (Base sequence) coding for amino acid sequence(s); (Universal) 3. The same codon/triplet (always) codes for the same amino acid;	3 (3 x AO1)	2. Ignore polypeptide chain/protein 3. Accept 3 bases for triplet/codon

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04.2	1. Correct mRNA sequence;	2 (2 x AO2)																															
	2. Correct amino acid sequence;																																
	<table><tr><td>DNA base sequence</td><td>G</td><td>C</td><td>T</td><td>C</td><td>T</td><td>A</td><td>C</td><td>A</td><td>G</td></tr><tr><td>mRNA base sequence</td><td>C</td><td>G</td><td>A</td><td>G</td><td>A</td><td>U</td><td>G</td><td>U</td><td>C</td></tr><tr><td>Amino acid sequence</td><td colspan="3">ARG</td><td colspan="3">ASP</td><td colspan="3">VAL</td></tr></table>			DNA base sequence	G	C	T	C	T	A	C	A	G	mRNA base sequence	C	G	A	G	A	U	G	U	C	Amino acid sequence	ARG			ASP			VAL		
	DNA base sequence			G	C	T	C	T	A	C	A	G																					
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04.3	1. (Deletion would cause) frame shift OR (Deletion) would change all following codons/triplets OR (Deletion) would change next codon/triplet from GAU (Asp) to AUG (Met); 2. (Substitution) would result in codon/triplet 1 being CGU/CGG/CGC; 3. (All) code for same amino acid/Arg/arginine OR Code is degenerate;	3 (3 x AO2)	1,2,3 Reject 'produce' amino acids once only